

Skills Summary

Vertices, Foci and Asymptotes of Ellipses and Hyperbolas

Use this reference while completing your worksheet or when studying.

Both curves are written with x^2 and y^2 over denominators. Which curve you have is decided by the sign between the two terms: a plus sign is an ellipse (a closed loop), a minus sign is a hyperbola (two branches). Which direction it runs is decided by the term that comes first, or is positive. The number under the transverse or major term is always a^2 .

Ellipse: vertices and foci from the equation

Rule: For $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with $a > b$, the major axis is horizontal: vertices $(\pm a, 0)$, co-vertices $(0, \pm b)$, foci $(\pm c, 0)$ with $c^2 = a^2 - b^2$.

Subtract for an ellipse: the foci sit *inside* the loop, so $c < a$. If the bigger denominator is under y^2 instead, everything turns a quarter-turn: vertices $(0, \pm a)$ and foci $(0, \pm c)$.

Example: Find the foci of $\frac{x^2}{100} + \frac{y^2}{36} = 1$.

Step 1. The bigger denominator, 100, is under x^2 , so this is a horizontal ellipse and the foci are on the x -axis — use $c^2 = a^2 - b^2$.

Step 2. $c^2 = 100 - 36 = 64$, so $c = \sqrt{64}$.

Foci at $(\pm c, 0)$ with $c = 8$

Try it: Find c for $\frac{x^2}{225} + \frac{y^2}{144} = 1$.

6 :pæp

Hyperbola: vertices and foci

Rule: For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ the branches open left and right: vertices $(\pm a, 0)$, foci $(\pm c, 0)$ with $c^2 = a^2 + b^2$.

Add for a hyperbola: the foci sit *beyond* the vertices, so $c > a$ always. The positive term names the direction — if the y^2 term is the positive one, the branches open up and down and a^2 is the number under y^2 .

Example: Find the vertices and foci of $\frac{x^2}{9} - \frac{y^2}{16} = 1$.

Step 1. The x^2 term is positive, so the branches open sideways and both vertices and foci are on the x -axis — use $c^2 = a^2 + b^2$.

Step 2. $a = \sqrt{9} = 3$, so the vertices are $(\pm 3, 0)$; and $c^2 = 9 + 16 = 25$.

Foci at $(\pm c, 0)$ with $c = 5$

Try it: Find c for $\frac{x^2}{64} - \frac{y^2}{36} = 1$.

01 :pæp

Hyperbola: slopes of the asymptotes

Rule: Draw the central rectangle: a each way along the transverse axis, b each way along the other. The asymptotes are its two diagonals extended.

Opening sideways: $y = \pm \frac{b}{a}x$. Opening up and down: $y = \pm \frac{a}{b}x$. Either way the number from under the *positive* term goes on the bottom when the curve opens sideways and on the top when it opens up and down. An ellipse has no asymptotes at all.

Example: Find the asymptote slopes of $\frac{x^2}{25} - \frac{y^2}{4} = 1$.

Step 1. The x^2 term is positive, so this hyperbola opens sideways and the slopes are $\pm \frac{b}{a}$ — rise b over run a on the central rectangle.

Step 2. $a = \sqrt{25} = 5$ and $b = \sqrt{4} = 2$, so the slope is $\frac{2}{5}$, and the asymptotes are $y = \pm \frac{2}{5}x$.
Slope = $\frac{2}{5}$

Try it: Find the asymptote slope of $\frac{x^2}{49} - \frac{y^2}{9} = 1$.

$\frac{2}{5}$:xpcəp

(!) Watch out: Do not read a as “the number under x^2 ”. For an ellipse a^2 is the *larger* denominator; for a hyperbola a^2 is the denominator under the *positive* term. Getting that backwards inverts every asymptote slope. And never take the square root of the whole equation: the denominators are a^2 and b^2 , so take roots of those numbers alone.